Review of Egor Jakimov

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The report presents the maximal inequality of Fefferman and Stein in a careful and rigorous way, and the proofs are generally easy to follow.

The duality between $\exp L$ and $L \log^+ L$ after Remark 5 is an important ingredient in the proof of (iii) and perhaps would fit better as a separate proposition. At the moment Chebyshev's inequality gets its own proposition while this is only listed as a fact.

Lemma 6 assumes f positive but is later applied to arbitrary f_k. Since $f_k^* = |f_k|^*$, it would be enough to assume $f_k \geq 0$ at the start of the proof of Theorem 7, or simply to drop the positivity assumption in Lemma 6.

I see why the order of the proof of Theorem 7 is necessary, since the case $p = r$ is needed for (ii), which in turn gives the remaining cases of (i). It would be easier to follow this non-linear structure with clearer markers and boundaries between the cases. For example, "This shows that (i) holds" on page 6 refers only to the case $p = r$ while I understood it to first claim proof of all cases of (i).

Moreover, vague references to the course and to Canvas could be replaced by references to the course literature, which would make the report more self-contained. The introduction could also say a few words about the motivation for these results. Finally, my opinion is that a short abstract and suitable title could add to the reading experience.